

Transmission Cooling



Inspection and Verification

NOTICE: When internal wear/damage occurs in the transmission, metal particles, clutch plate material and band material can travel into the torque converter, the transmission fluid cooler tubes, the in-tank transmission fluid cooler and the auxiliary transmission fluid cooler, if so equipped. These contaminants are a major cause of recurring transmission concerns. To prevent future concerns, remove these contaminants from the cooling system before placing the transmission back into use.

1. Visually inspect for obvious signs of mechanical damage, incorrect component installation and system leaks. Repair as necessary.
- Install a new or remanufactured radiator when transmission fluid is found leaking from the in-tank transmission fluid cooler. Refer to Section 303-03.

• Install a new auxiliary transmission fluid cooler if there is leakage from the auxiliary transmission fluid cooler.
2. If the fault is not visually evident, GO to [Symptom Chart — Transmission Cooling](#) or GO to [Symptom Chart — NVH](#).

Symptom Chart — Transmission Cooling

Symptom Chart — Transmission Cooling		
Condition	Possible Causes	Action
• Transmission overheating	• Excessive vehicle or towing loads, severe vehicle use • Incorrect transmission fluid level • Transmission fluid condition • Incorrect transmission fluid type • Transmission cooling system damaged, blocked, restricted or installed incorrectly • System leaks • Engine overheating • Powertrain control system electrical inputs/outputs, vehicle	• REFER to the Vehicle Specification Manual for load and GVW information. REFER to the Owner's Literature. • CHECK and if necessary, ADJUST the transmission fluid to the correct level. • CHECK transmission fluid condition. • REFER to Specifications in this section. • CARRY OUT the Transmission Fluid Cooler Backflushing and Cleaning. REFER to Section 307-01. • INSPECT for leakage. • REFER to Section 303-03. • CARRY OUT On-Board Diagnostic (OBD) test. REFER to

	wiring harnesses, PCM, Torque Converter Clutch (TCC) solenoid Internal transmission concerns	the Powertrain Control/Emissions Diagnosis (PC/ED) manual. REPAIR as necessary.
Transmission fluid leak at radiator in-tank cooler	Transmission fluid cooler tubes-to-transmission fluid cooler fitting	REFER to the Component Test in this section.

Symptom Chart — NVH

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Condition	Possible Causes	Action
NOTE: <i>NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.</i>		
Vibration — a high frequency (20-80 Hz) that is felt through the seat or selector lever. Changes with engine speed	Transmission fluid cooler tubes grounded out	CHECK the transmission fluid cooler tubes. REPAIR as necessary.

Component Test

Transmission Fluid Cooler Leaks at Radiator

Transmission fluid may leak between the radiator transmission fluid cooler and the transmission fluid cooler fitting (not the transmission fluid cooler tube into the transmission fluid cooler fitting), which may result in a residue of transmission fluid on the radiator tank around the transmission fluid cooler fitting. Insufficient thread sealer on the transmission fluid cooler fitting may cause this.